Computer Networking

Password Security and Password Breaking



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1| What is Router Security

Router password setup is the process of configuring passwords on a router to protect it from unauthorized access and ensure network security.



2| Why it is important?

Key Points

- It restricts access to the router
- Only authorized users can log in
- Prevents unwanted configuration changes
- Protects network data and devices
- Increases overall network security

Example

- A router **without a password**
  → Anyone can log in and change settings (IP, WiFi, security)
- A router **with a password**
  → Only the admin can access and control the router

3. Purpose of Router Password Setup

The main purpose of router password setup is to protect the router from unauthorized access and keep the network secure.

Router password setup is to protect the router and the entire network from unauthorized access. When a password is configured, only authorized users such as the network administrator—can log in and control the router. This ensures that important settings like IP configuration, routing, DNS, and security options cannot be changed by outsiders.
It also plays a crucial role in maintaining network stability. Without a password, anyone could enter the router and misconfigure settings, which might disconnect users from the internet or slow down the network. In addition, password protection helps prevent cyber attacks, data theft, and misuse of network resources. It keeps sensitive information safe and ensures that only trusted users can manage or monitor network activity.

For example, imagine a small office network. If the router has no password, any employee or even an outsider connected to the network can log in and change the Wi-Fi password or block internet access for others. They could also redirect traffic or create security vulnerabilities. On the other hand, if the router is protected with a strong password, only the network administrator can access it and make changes, keeping the network secure and running smoothly.



4. Password Security Types

Router password security is divided into different types based on **where and how the password is used** to protect access.

- **Console Password**

This password is used when someone tries to access the router physically using a console cable. It protects direct access to the router. If a person connects a laptop to the router using a console cable, they must enter the console password to log in.

- **Enable Password / Enable Secret**

This is used to protect **privileged (admin) mode** in the router.

- **Enable Password** → less secure (stored in plain text)
- **Enable Secret** → more secure (encrypted by MD 5)

A user logs into the router but cannot enter admin mode unless they provide the enable password or secret.

- **VTY Password (Telnet/SSH)**

This password is used for **remote access** to the router over the network. Example:
If someone tries to access the router using Telnet or SSH from another PC, they must enter the VTY password.

- **Auxiliary (AUX) Password**

This is used for **remote access via modem connection** (less common today).
Example: An administrator connects to the router using a modem and must enter the AUX password.

Router password security types are used to protect different access methods such as **physical access, remote access, and administrative control**, ensuring that only authorized users can manage the router.

Imagine a company network:

- An engineer connects directly using a cable → needs **console password**
- Then tries to enter admin mode → needs **enable secret**
- Another admin connects from another computer → needs **VTY password**

5. Router Password Configuration

Router(config)# line console 0
Router(config-line)#password cisco ← sets console (user mode) password
Router(config-line)#login ← to call the authentication Page
Router(config-line)#exit
Router(config)# enable password ccna ← sets privileged (enable) mode password
Router# copy running-config startup-config or write memory ← saves configuration to startup

Router(config)# line vty 0 2 ← how many pc can connect via Remote login
Router(config-vty)#password 1234 ← sets vty password as 1234
Router(config-vty)#login ← to call the authentication Page
Router(config-vty)#exit

How to Break password on Router:

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- 1) First switch off the router and then on it. After that press
- [Ctrl + Shift + 6 + c] or [Ctrl + Pause Break] ← to interrupt the router boot process.
- Router has 2 Register value for breaking password:
  a. 0x2102 = Default mode (reload/restart)
  b. 0x2142 = Reset mode
- Rommom = router OS
- 2) Rommon1> confreg 0x2142
- 3) Rommon2> reset
- 4) Router> en
- 5) Router# show run → nothing to show
- 6) Router# copy startup-config running-config
- 7) Router# show run → check password
- 8) Router(config)# config-register 0x2102
- 9) Router(config)# exit
- 10) Router# reload

How to Remove password on Router:

Router(config)# line console 0
Router(config-line)# no password
Router(config-line)# exit
Router(config)# no enable password / no enable secret
Router(config)# line vty 0 2
Router(config-vty)# no password
Router(config-vty)# exit



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